

Module 04: Cognition in Embodied Cognition — Thinking on the Move

Learning Objectives

1. Define **mobile cognition** as the cognitive processes that emerge from and are shaped by active movement through space.
2. Analyze how **spatial reasoning, environmental affordance assessment, and route planning** are enhanced by physical engagement with the environment.
3. Apply the Active Inference framework to explain why walking aids thinking, and why physical environments shape cognitive processes.

Introduction

Some of the best thinking happens in motion. Darwin walked his "Sandwalk" daily to think through evolutionary theory. Aristotle taught while walking (the Peripatetic school). Steve Jobs was famous for "walking meetings." This is not coincidence — movement through the world actively engages cognitive systems that support creative and analytical thinking.

Key Concepts

1. Walking and Creative Cognition

Research demonstrates that walking boosts creative thinking (Oppezzo & Schwartz, 2014):

- Walking increased creative output by 60% compared to sitting — for both indoor treadmill and outdoor walking
- The effect persists briefly after walking stops — the primed generative model continues to run in its enriched mode
- Active Inference interpretation: Walking generates a continuous stream of low-level prediction errors (terrain, environment, proprioceptive variations) that loosens the generative model's precision structure — making it more "exploratory" and open to

novel associations. This loosened precision *generalizes* to abstract thought, producing creative divergent thinking.

2. Environmental Affordance Assessment

Moving through the world requires continuous **cognitive assessment** of environmental affordances:

- "Can I fit through that gap?" requires real-time body-environment scaling computation (Warren & Whang, 1987: subjects accurately judged passability at $1.16\times$ shoulder width)
- "Is that surface safe to walk on?" requires material assessment, slope estimation, and friction prediction
- "Can I reach that object?" requires evaluating the reach envelope relative to body position and obstacles

These assessments are not separate from movement — they are the **cognitive content** of movement, the ongoing inference that makes locomotion possible.

3. Cognitive Mapping During Movement

The cognitive map is not built from static observation — it is constructed through active movement:

- **Path integration:** The cognitive system continuously updates estimated position using self-motion signals — a form of dead reckoning that relies on the vestibular-proprioceptive generative model
- **Landmark-based updating:** When recognized landmarks are encountered, the position estimate is corrected — analogous to the Kalman filter's observation update step
- **Exploration strategies:** The agent's movement pattern determines what parts of the environment are mapped — systematic searchers build complete maps; habitual route-takers build detailed knowledge of corridors between familiar landmarks

4. Embodied Decision-Making in Action

Decisions made during movement are constrained by the temporal and physical demands of ongoing action:

- A pedestrian approaching a crosswalk must decide to cross or wait within seconds — integrating traffic speed, gap size, walk speed, and risk tolerance into a time-critical embodied decision
- A rock climber mid-route must choose the next handhold from partial information while managing muscle fatigue, grip strength, and fall consequence — a multi-factor decision made under time pressure with somatic inputs
- These decisions engage the body's fast inference systems (somatic markers, motor imagery, proprioceptive simulation) rather than deliberate analytical reasoning

Applications

- **Walking meetings in organizations:** Companies that adopt walking meetings leverage the cognitive benefits of movement — the loosened precision structure that promotes creative thinking, combined with the shared locomotor experience that promotes social alignment (pace matching, co-navigation) and reduces hierarchical barriers.
- **Environmental design for cognition:** University campuses, hospital corridors, and office building layouts can be designed to enhance cognitive function — walking paths that offer visual variety (varied prediction errors), natural elements (biophilic restoration of cognitive resources), and comfortable waypoints (controlled environmental affordances) support better thinking and decision-making.

Conclusion

Cognition during movement is enriched by the sensorimotor engagement that walking provides — creative priming, affordance assessment, cognitive mapping, and embodied decision-making all benefit from active physical engagement with the environment. The next module examines embodied action during world-engaged movement.